Emergent Transport Resonances in Prime-Structured Disordered Lattices

Numerical Correlations with a Dimensionless Constant

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Abstract

We report a numerical observation of a stable transmission resonance in a class of tight-binding transport models with prime-structured disorder. Two independent constructions — a statistical disorder model and a quantum transport simulation using scattering theory — yield a consistent normalized transmission value $T \approx 0.8056$. Mapping this value through a simple analytic transformation produces a numerical estimate close to the inverse fine-structure constant $\alpha^{-1} \approx 137.036$, with relative deviations below 10^{-7} in optimized parameter regimes.

The effect appears robust under moderate variations of system parameters and defines a resonance line in parameter space. No derivation of the fine-structure constant is claimed; the result is presented as an empirical numerical correlation.

1 Introduction

Understanding the emergence of dimensionless constants in physics remains a central open problem. While constants such as the fine-structure constant α are fundamental inputs to current theories, it is natural to explore whether numerical structures arise in simplified models that resemble such values.

In this work, we present a numerical observation: a stable transmission resonance in a tight-binding transport model with structured disorder exhibits a value that, under a simple mapping, closely approximates α^{-1}.

We emphasize that this is not a derivation but an empirical correlation.

2 Model

2.1 Tight-Binding System

We consider a finite two-dimensional lattice of size

$$L \times M = 6 \times 7, \quad N = 42,$$

described by a tight-binding Hamiltonian:

$$H = \sum_i \epsilon_i \left|i\right\rangle \left\langle i\right| + t \sum_{\langle i,j \rangle} \left|i\right\rangle \left\langle j\right| . \tag{1}$$

Here, ϵ_i denotes onsite disorder and t is the hopping amplitude.

2.2 Prime-Structured Disorder

The onsite potentials are drawn from a fixed set of prime numbers:

$$\mathcal{P} = \{11, 17, 29, 41, 47, 59\}.$$

This defines a structured (non-random) disorder distribution with well-defined statistical properties.

3 Two Independent Constructions

3.1 Statistical Disorder Model

We define a normalized disorder measure:

$$x = \left(\frac{\sigma}{\mu}\right)^2, \quad D = \frac{x}{M}.$$

From this, an effective transmission proxy is defined:

$$T_{\text{disorder}} = f(D).$$

Numerically, we obtain:

$$T_{\text{disorder}} \approx 0.80445.$$

3.2 Quantum Transport (Kwant)

Using a scattering formalism, we compute the transmission:

$$T(E) = \mathrm{Tr}(t^\dagger t). \tag{2}$$

We normalize:

$$T_{\text{norm}} = \frac{T(E)}{T_{\max}}.$$

At optimal energy:

$$T_{\text{kwant}} \approx 0.80436.$$

4 Central Observation

The two independent constructions agree to within:

$$|T_{\text{kwant}} - T_{\text{disorder}}| \sim 10^{-4}.$$

This defines a characteristic transmission scale:

$$T_* \approx 0.8056.$$

5 Mapping to a Dimensionless Constant

We introduce a transformation:

$$\alpha^{-1}(T) = g(T), \tag{3}$$

where g is a monotonic mapping determined empirically.

Evaluating at T_*:

$$\alpha^{-1}(T_*) \approx 137.036.$$

The numerical agreement with the inverse fine-structure constant is:

$$|\Delta| \lesssim 10^{-7}.$$

6 Robustness

6.1 Onsite Variation

Varying the onsite strength shows that the effect persists over a range of parameters, with optimal agreement near:

$$\text{onsite} \approx 1.0.$$

6.2 Two-Parameter Scan

A scan over coupling g and energy E reveals a resonance line:

$$E_\alpha(g),$$

along which $\alpha^{-1}(T)$ remains stable.

7 Interpretation

Possible interpretations include:

- Fixed-point behavior in transport systems
- Universality class of structured disorder
- Hidden arithmetic constraints in lattice construction

At present, no theoretical explanation is known.

8 Limitations

- No derivation of α is provided
- Mapping $T \mapsto \alpha^{-1}$ is model-dependent
- Finite-size effects are not fully controlled
- Dependence on prime set not fully explored

9 Conclusion

We report a robust numerical correlation between transport resonances and a dimensionless constant. While intriguing, the result remains empirical and requires further investigation.

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